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Area and Perimeter

Five area formulas, then split any shape into them. Perimeter: walk round the edge and add every length.

THE TOPIC ON ONE MAP

Area and Perimeter

01 Area of standard shapes

trapezium

$$\frac{1}{2}ab \sin C$$

parallelogram

02 Compound shapes

split or subtract

how many tins?

03 Area with circles

semicircles on a square

shaded region

04 Working backwards and algebra

find x from the area

find the perimeter

THE FORMULAS Know these five by heart:

ZONE 1

Rectangle



$$bh$$

ZONE 2

Triangle



$$\frac{1}{2}bh$$

ZONE 3

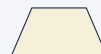
Parallelogram



$$bh$$

ZONE 4

Trapezium



$$\frac{1}{2}(a + b)h$$

ZONE 5

Circle



$$\pi r^2$$

01 Area of Standard Shapes

Pick the formula from the shape. The height is always **perpendicular** to the base — never the slanted side.

WHEN YOU SEE → YOU DO

“trapezium”	→ $\frac{1}{2}(a + b)h$ with the parallel sides
two sides and the angle between them	→ $\frac{1}{2}ab \sin C$
“one line of symmetry” trapezium	→ the two overhangs are equal
area in m^2 , price per hectare	→ 1 hectare = 10 000 m^2

WORKED MODEL • JUNE 2025 P1H

Isosceles trapezium: $AD = 12$, angle $D = 60^\circ$, $DC = 47$. Find the area.

1 height = $12 \sin 60^\circ = 10.39$

perpendicular height

2 overhang = $12 \cos 60^\circ = 6$, top = $47 - 12 = 35$

symmetry

ANSWER $\frac{1}{2}(35 + 47)(10.39) = 426 \text{ cm}^2$

× **TRAP** Using the slanted side as the height. The height meets the base at 90° .

IDEA 01

Practise: Area of Standard Shapes

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Trapezium

CH 4 • TOPIC 5 2 marks

A trapezium has parallel sides 9 cm and 15 cm and height 6 cm. Work out its area.

SELECTED PRACTICE**Q2** PRACTICE Practice • Medium

CH 4 • TOPIC 5 2 marks

2 Here is a trapezium drawn on a centimetre grid.



On the grid, draw a triangle equal in area to this trapezium.

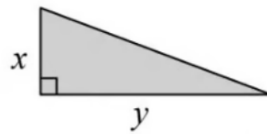
(2 marks)

Q3**PRACTICE**

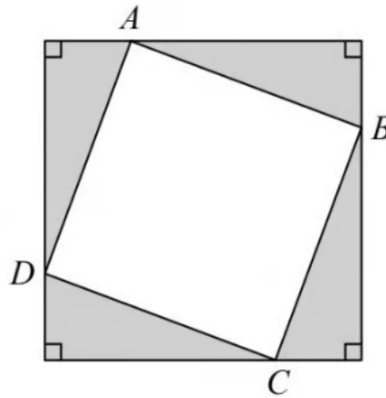
Practice · Hard

CH 4 · TOPIC 5 3 marks

3 Here is a right-angled triangle.



Four of these triangles are joined to enclose the square $ABCD$ as shown below.



Show that the area of the square $ABCD$ is $x^2 + y^2$.

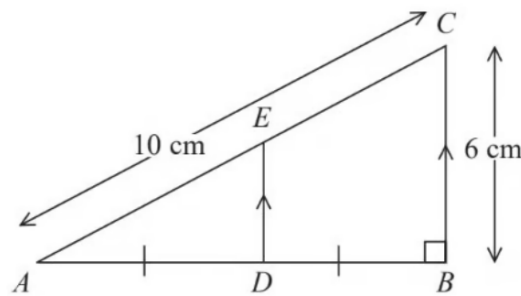
(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 4 · TOPIC 5 4 marks

9

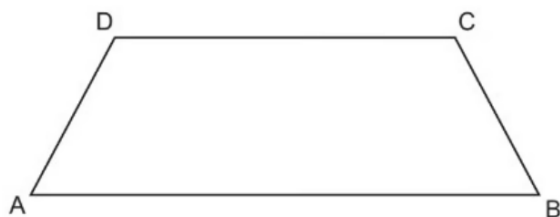
Diagram **NOT**
accurately drawn ADB and AEC are straight lines. DE is parallel to BC .Angle $ABC = 90^\circ$ $AC = 10$ cm. $BC = 6$ cm. D is the midpoint of AB .Work out the area of trapezium $BCED$.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 5 7 marks

14 ABCD is a trapezium.

Not to scale

The perimeter of the trapezium is 56 cm.

The ratio $AD : AB : DC : BC = 5 : 12 : 6 : 5$.

Calculate the area of the trapezium.

Show your working.

.....cm²**(7 marks)**

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q11

CH 4 · TOPIC 5 6 marks

- 11 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

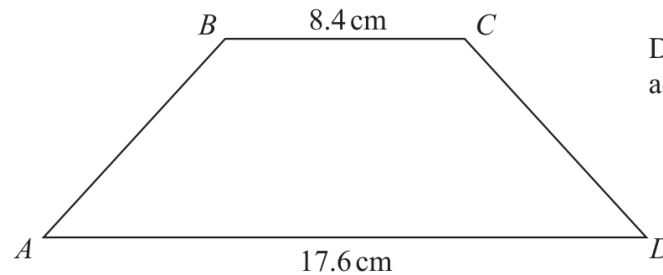


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

The trapezium has area 179.4 cm^2

Work out the size of angle ABC .

Give your answer correct to 1 decimal place.

o

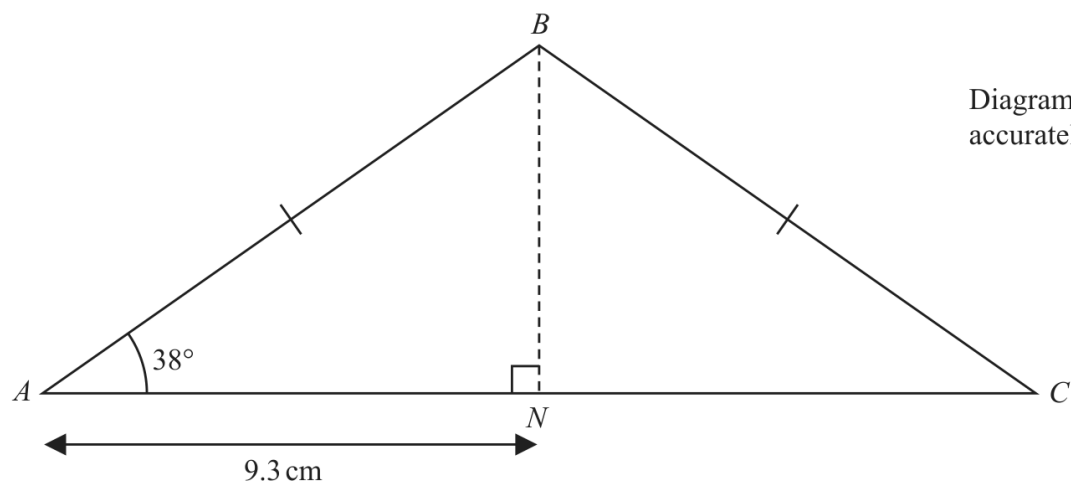
(Total for Question 11 is 6 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q10

CH 4 · TOPIC 5 4 marks

10 ABC is an isosceles triangle with $BA = BC$.



N is the point on AC such that $AN = 9.3$ cm and BN is perpendicular to AC .

Work out the perimeter of triangle ABC .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 4 marks)

Q8 **PAST PAPER** Edexcel 4MA1 · June 2023 · Paper 2HR · Q5**CH 4 · TOPIC 5** **4 marks**

- 5 A field is in the shape of a trapezium.

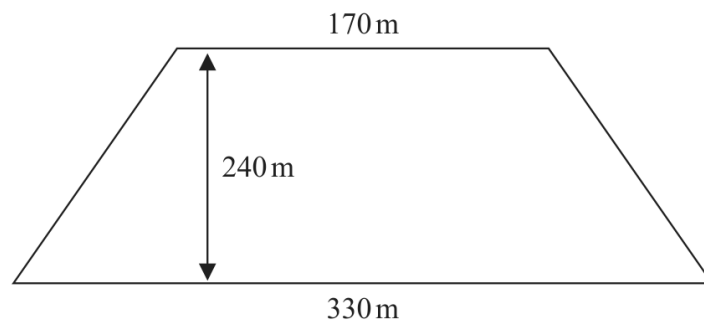


Diagram **NOT**
accurately drawn

The field is sold for a price of \$49 650

Given that 1 hectare = 10 000 m²

work out the average price of the field per hectare.

\$.....

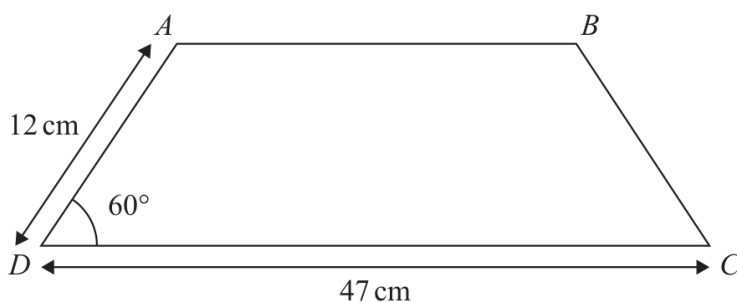
(Total for Question 5 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q9

CH 4 · TOPIC 5 5 marks

9



$ABCD$ is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^2

(Total for Question 9 is 5 marks)

02 Compound Shapes

Split the shape into rectangles, triangles and trapeziums — or take a small shape away from a big one. Work out any missing lengths first.

WHEN YOU SEE → YOU DO

- | | |
|-------------------------------|---|
| an L-shape or a stepped shape | → split into rectangles; missing lengths by subtraction |
| a shape with a piece cut out | → big area – small area |
| “how many tins / packs” | → area ÷ coverage, round up |
| “is there enough money?” | → number of tins × price after discount |

Missing lengths

Split or subtract

Add the areas

Answer the story

WORKED MODEL • JANUARY 2021 P1HR

A pentagon face: rectangle 10 m by 6 m and an isosceles triangle with sides 7 m. Paint covers 16 m^2 per tin. Least number of tins?

1 triangle height = $\sqrt{7^2 - 5^2} = \sqrt{24}$

Pythagoras

2 $60 + \frac{1}{2}(10)\sqrt{24} = 84.5 \text{ m}^2$

add the parts

ANSWER $84.5 \div 16 = 5.3 \Rightarrow 6 \text{ tins}$

IDEA 02

Practise: Compound Shapes

My questions first, then selected practice, then real past papers.

Q10**DR ESLAM**

Split it

CH 4 · TOPIC 5

3 marks

An L-shape is a 10 m by 8 m rectangle with a 4 m by 3 m corner removed. Find its area and perimeter.

SELECTED PRACTICE

Q11 PRACTICE Practice · Medium

CH 4 · TOPIC 5 4 marks

- 1 The diagram shows the plan of a small field.

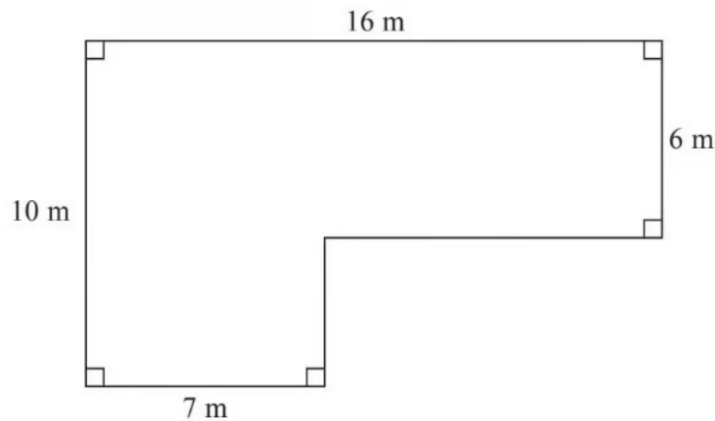


Diagram **NOT**
accurately drawn

Kevin is going to keep some pigs in the field.
Each pig needs an area of 36 square metres.

Work out the greatest number of pigs Kevin can keep in the field.

(4 marks)

Q12

PRACTICE

Practice · Medium

CH 4 · TOPIC 5

4 marks

14 Here is a hexagon $ABCDEF$.

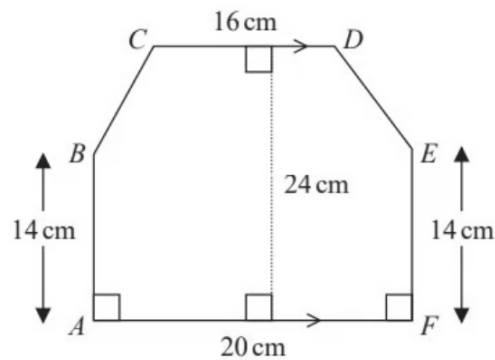


Diagram **NOT**
accurately drawn

CD is parallel to AF .

Work out the area of hexagon $ABCDEF$.

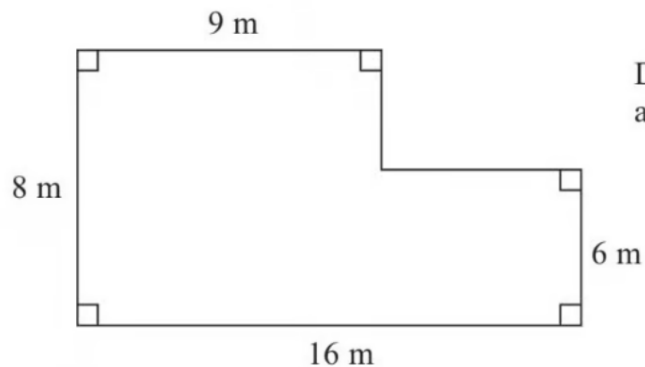
..... cm^2
(4 marks)

Q13**PRACTICE**

Practice · Hard

CH 4 · TOPIC 5 5 marks

12 The diagram shows the floor of a village hall.



The caretaker needs to polish the floor.

One tin of polish normally costs £19

One tin of polish covers 12 m^2 of floor.

There is a discount of 30% off the cost of the polish.

The caretaker has £130

Has the caretaker got enough money to buy the polish for the floor?

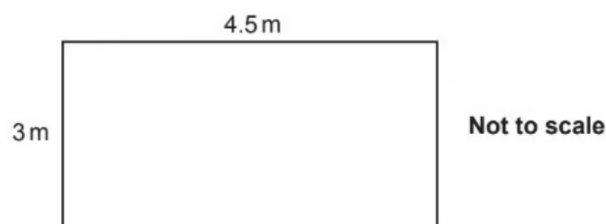
You must show all your working.

(5 marks)

Q14 **PRACTICE** Practice · Very Hard

CH 4 · TOPIC 5 6 marks

16 Here is the floor plan of a rectangular room.



Tim buys carpet tiles for this room.

Each tile is a square measuring 50cm by 50cm.

The tiles are only sold in packs of ten.

Each pack costs £20.

Tim pays for fitting at a rate of £7.50 per square metre, with any fraction of a square metre rounded up.

Work out the **total** cost of the tiles and fitting.

£

(6 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q22

CH 4 · TOPIC 5 6 marks

22 The diagram shows a regular octagon $ABCDEFGH$.

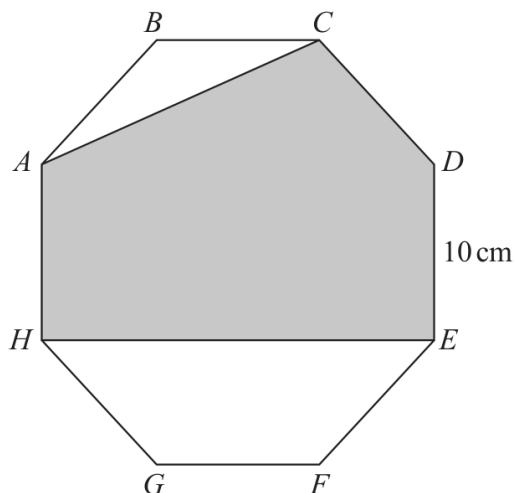


Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm .

Find the area of the shaded region $ACDEH$.

Give your answer correct to the nearest cm^2

..... cm^2

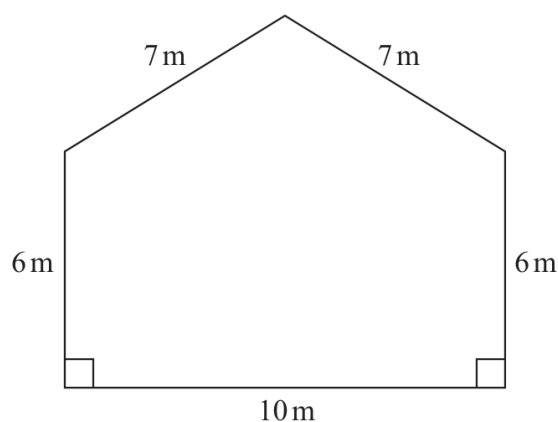
(Total for Question 22 is 6 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q10

CH 4 · TOPIC 5 5 marks

- 10 The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.



Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

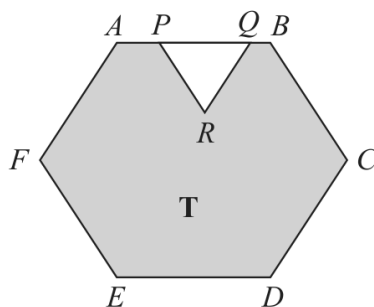
The paint in each tin of paint Omondi is going to buy will cover 16 m^2 of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

(Total for Question 10 is 5 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q20

CH 4 · TOPIC 5 4 marks**20**Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

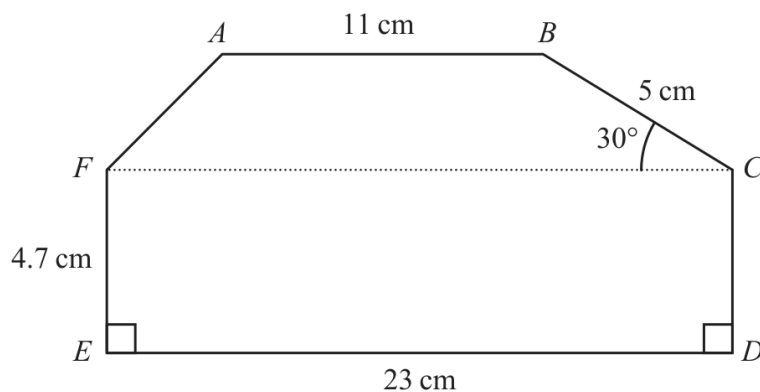
work out the length of PQ .

..... cm

(Total for Question 20 is 4 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q10

CH 4 · TOPIC 5 5 marks10 The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawnAngle $BCF = 30^\circ$ AB , FC and ED are parallel.Calculate the area of $ABCDEF$

Show your working clearly.

..... cm^2

(Total for Question 10 is 5 marks)

03 Area with Circles

Circle πr^2 and $2\pi r$; a semicircle is half. For a shaded region: whole shape minus the unshaded part.

WHEN YOU SEE → YOU DO

“AC is a diameter”

→ radius = $\frac{1}{2}AC$; angle in a semicircle 90°

semicircles on the sides of a square

→ square + 4 half-circles

“perimeter of the shaded shape”

→ curved part πr + the straight parts

“give your answer to 3 s.f.”

→ keep π until the last line

WORKED MODEL · JANUARY 2021 P2HR

Rectangle 12×6 with a semicircle of diameter 12 inside. Shaded region?

1 $12 \times 6 = 72$

rectangle

2 $\frac{1}{2}\pi(6)^2 = 18\pi$

semicircle

ANSWER $72 - 18\pi = 15.5 \text{ cm}^2$

IDEA 03

Practise: Area with Circles

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM Semicircle

CH 4 · TOPIC 5 3 marks

A shape is a 10 cm square with a semicircle on one side. Find its perimeter to 3 s.f.

SELECTED PRACTICE

Q20

PRACTICE

Practice · Easy

CH 4 · TOPIC 5 4 marks

- 1 The diagram shows a circle and a trapezium.

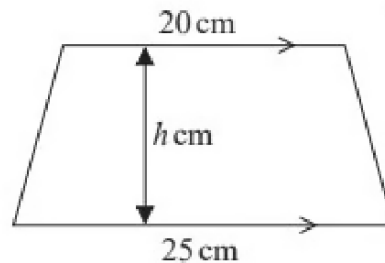
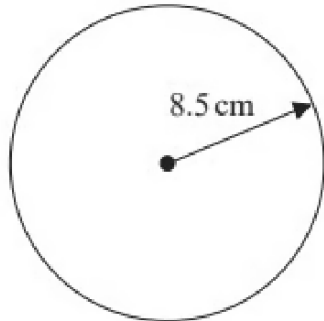


Diagram **NOT**
accurately drawn

The height of the trapezium is h cm.

The area of the circle is equal to the area of the trapezium.

Work out the value of h .

Give your answer correct to 1 decimal place.

(4 marks)

Q21

PRACTICE

Practice · Medium

CH 4 · TOPIC 55 marks

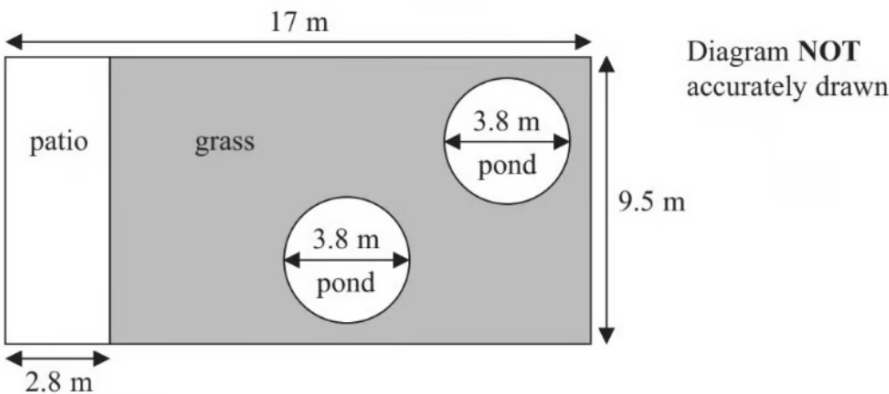
PRACTISE · 4.5

7 Mr Weaver's garden is in the shape of a rectangle.

In the garden

	there is a patio in the shape of a rectangle
and	two ponds in the shape of circles with diameter 3.8 m.

The rest of the garden is grass.



Mr Weaver is going to spread fertiliser over all the grass.
One box of fertiliser will cover 25 m^2 of grass.

How many boxes of fertiliser does Mr Weaver need?
You must show your working.

(5 marks)

Q22**PRACTICE**

Practice · Hard

CH 4 · TOPIC 5 4 marks

- 19** The diagram shows equilateral triangle ABC with sides of length 10 cm. A circle is drawn inside the triangle.

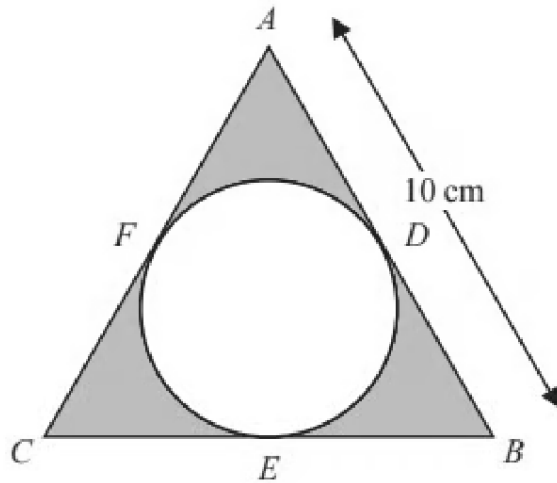


Diagram **NOT**
accurately drawn

D , E and F are points on the circle

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

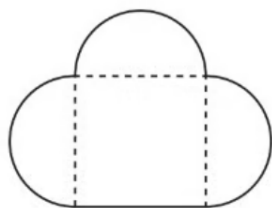
(4 marks)

Q23**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 5 6 marks

- 13** The diagram shows Jane's lawn. It is in the shape of a square of side 36m and three semi-circles.



Not to scale

She is going to spread fertiliser on the lawn at a rate of 30g per square metre.

The fertiliser is only sold in 10kg bags costing £15.80 each.

Calculate the cost of buying the bags of fertiliser for her lawn. You must show all your working.

£

(6 marks)

PAST PAPERS

Q24 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1HR · Q5

CH 4 · TOPIC 5 3 marks

- 5 The diagram shows Yuen's garden.

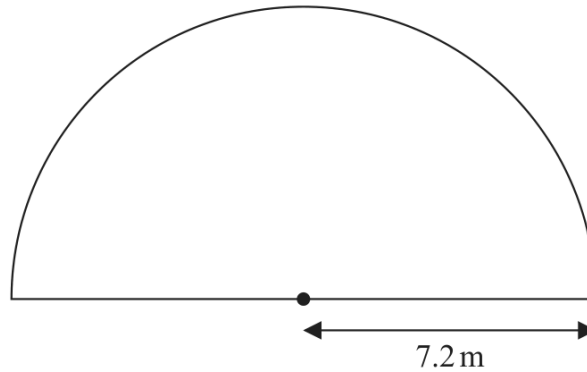


Diagram **NOT**
accurately drawn

The garden is in the shape of a semicircle of radius 7.2 m.
Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.
Each box of grass seed contains enough seed to cover 6 m^2 of the garden.

Has Yuen enough grass seed for his garden?
Show your working clearly.

(Total for Question 5 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q11

CH 4 · TOPIC 5 5 marks

- 11 The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

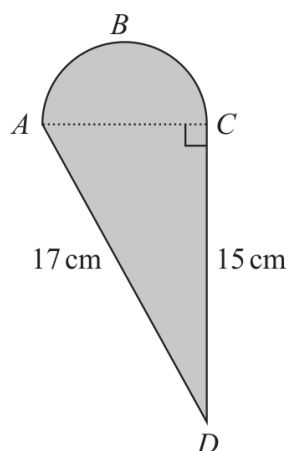


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

Work out the perimeter of the shaded shape.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 11 is 5 marks)

Q26**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q2

CH 4 · TOPIC 5 3 marks

- 2 The diagram shows a rectangle $ABCD$ and a semicircle with diameter AB where $AB = 12$ cm. The point E lies on DC and also on the semicircle.

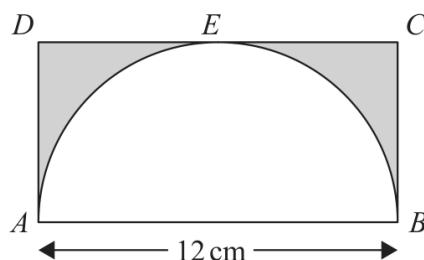


Diagram **NOT**
accurately drawn

Work out the area of the shaded region.

Give your answer correct to 3 significant figures.

..... cm²

(Total for Question 2 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q8

CH 4 · TOPIC 5 5 marks

8 A , B and C are points on a circle with centre O .

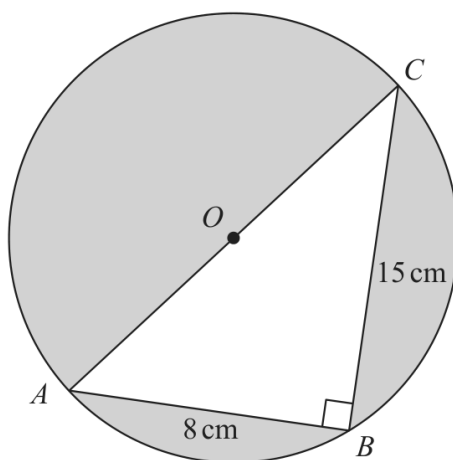


Diagram **NOT**
accurately drawn

AOC is a diameter of the circle.

$AB = 8\text{ cm}$ $BC = 15\text{ cm}$

Angle $ABC = 90^\circ$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 8 is 5 marks)

Q28**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 1H · Q4

CH 4 · TOPIC 5 5 marks

- 4 The diagram shows the front of a wooden door with a semicircular glass window.

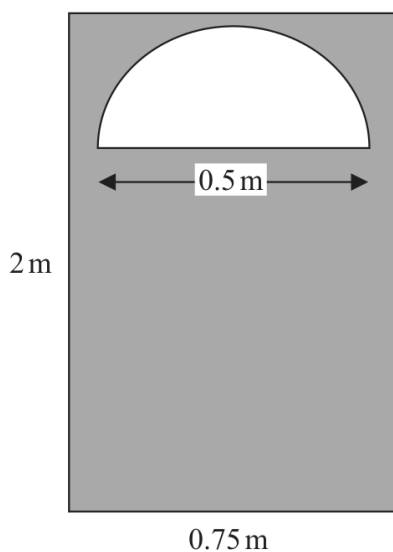


Diagram **NOT**
accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

(Total for Question 4 is 5 marks)

Q29**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q5

CH 4 · TOPIC 5 4 marks

- 5 The diagram shows a shape made from a square $ABCD$ and 4 identical semicircles.

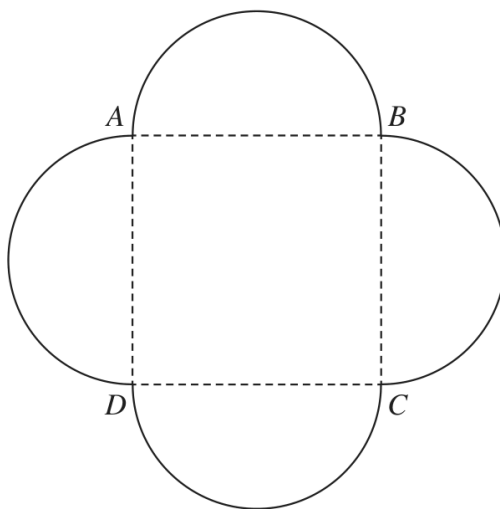


Diagram **NOT**
accurately drawn

As shown in the diagram, the semicircles have AB , BC , CD and DA as diameters.

The area of the square is 36 cm^2

Calculate the total area of the shape.

Give your answer correct to one decimal place.

..... cm^2

(Total for Question 5 is 4 marks)

04 Working Backwards and Algebra

Area given, length wanted: write the area in terms of x , set it equal, solve. Check that every length stays positive.

WHEN YOU SEE → YOU DO

“the shape has area 129 cm^2 — find x ”

→ area expression = 129

“perimeter of the hexagon = perimeter of the triangle”

→ two expressions equal

a quadratic in x

→ reject the answer that makes a length negative / too big

“work out the perimeter” after finding a length

→ Pythagoras for the slanted side

WORKED MODEL · MAY 2024 P1HR

Right trapezium: $AD = 15$, $DC = 14$, area 360. Find the perimeter.

1 $\frac{1}{2}(AB + 14)(15) = 360 \Rightarrow AB = 33$

area equation

2 $BC = \sqrt{15^2 + 19^2} = 24.2$

Pythagoras

ANSWER $33 + 15 + 14 + 24.2 = 86.2 \text{ cm}$

IDEA 04

Practise: Working Backwards and Algebra

My questions first, then selected practice, then real past papers.

Q30 DR ESLAM Backwards

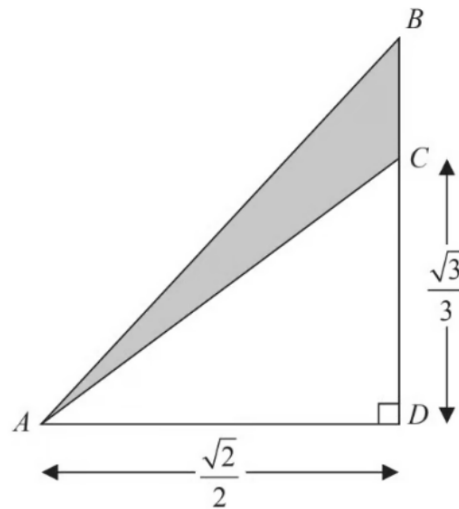
CH 4 · TOPIC 5 3 marks

A triangle has area 42 cm^2 and base 12 cm. Find its perpendicular height.

SELECTED PRACTICE

Q31 PRACTICE Practice · Medium

CH 4 · TOPIC 5 3 marks

11 ABD is a right angled triangle.Diagram **NOT**
accurately drawn

All measurements are given in centimetres.

 C is the point on BD such that $CD = \frac{\sqrt{3}}{3}$

$$AD = BD = \frac{\sqrt{2}}{2}$$

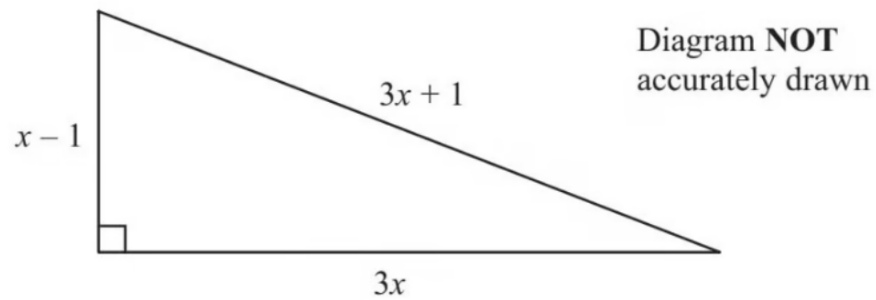
Work out the exact area, in cm^2 , of the shaded region.**(3 marks)**

Q32**PRACTICE**

Practice · Hard

CH 4 · TOPIC 5 4 marks

- 1 The diagram shows a triangle.



In the diagram, all the measurements are in metres.

The perimeter of the triangle is 56 m.

The area of the triangle is $A \text{ m}^2$.

Work out the value of A .

(4 marks)

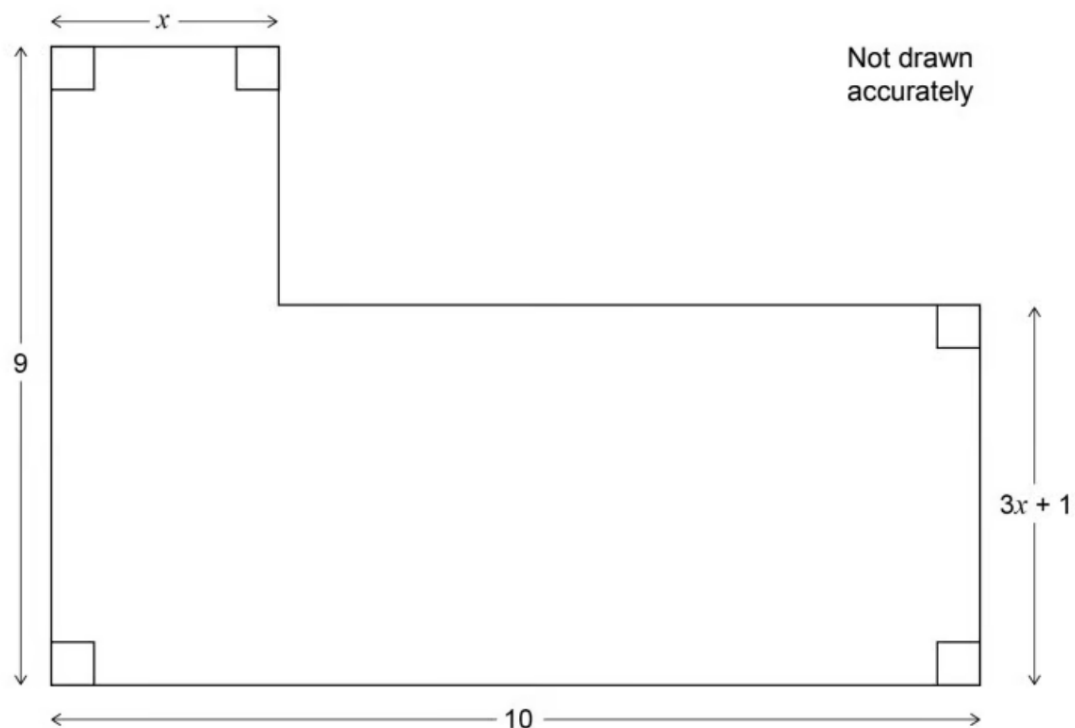
Q33**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 5 6 marks

11 Here is an L-shape.

All dimensions are in centimetres.

The area of the L-shape is 65 cm^2 Work out the value of x .**(6 marks)**

Q34**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 5 3 marks

- 17** The lengths of the sides of two squares are integers, when measured in cm.
The difference between the areas of the two squares is 36cm^2 .

Find the lengths of the sides of the two squares.

..... cm

..... cm

(3 marks)

PAST PAPERS

Q35 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1HR · Q10

CH 4 · TOPIC 5 5 marks

10 The diagram shows a regular hexagon, $ABCDEF$, and an isosceles triangle, GHI .

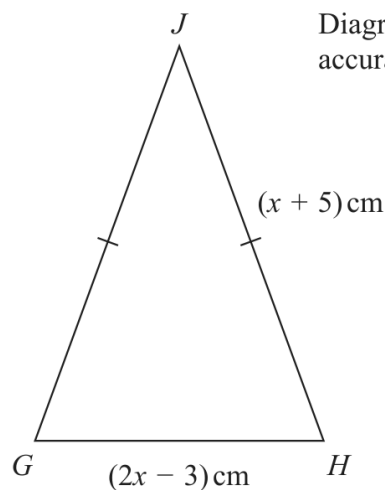
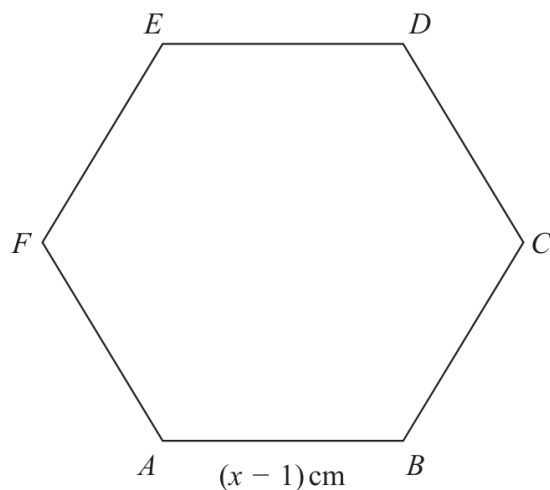


Diagram **NOT**
accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.
Show clear algebraic working.

..... cm

(Total for Question 10 is 5 marks)

Q36 **PAST PAPER** Edexcel 4MA1 · January 2020 · Paper 1H · Q3**CH 4 · TOPIC 5** **4 marks**

- 3 The diagram shows a shape.

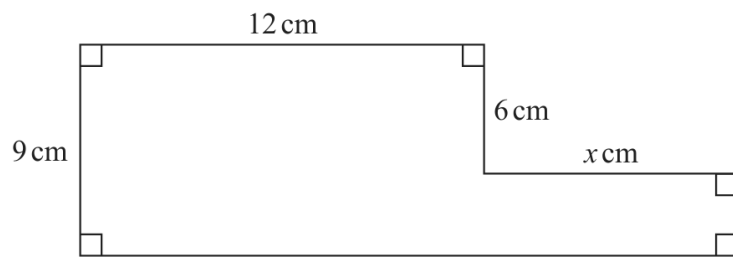


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

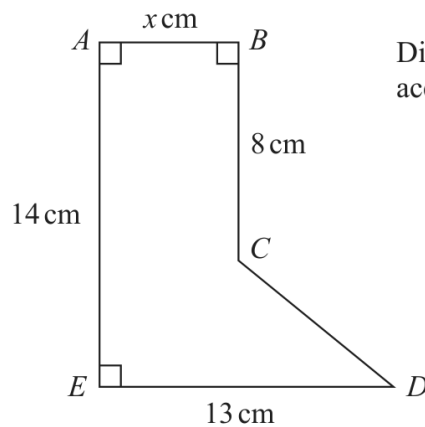
$x = \dots\dots\dots$

(Total for Question 3 is 4 marks)

Q37 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 1H · Q3

CH 4 · TOPIC 5 4 marks

3

Diagram **NOT**
accurately drawnThe diagram shows the shape $ABCDE$.The area of the shape is 91.8 cm^2 Work out the value of x . $x = \dots\dots\dots$

(Total for Question 3 is 4 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q11

CH 4 · TOPIC 5 6 marks

11 The diagram shows a quadrilateral $ABCD$

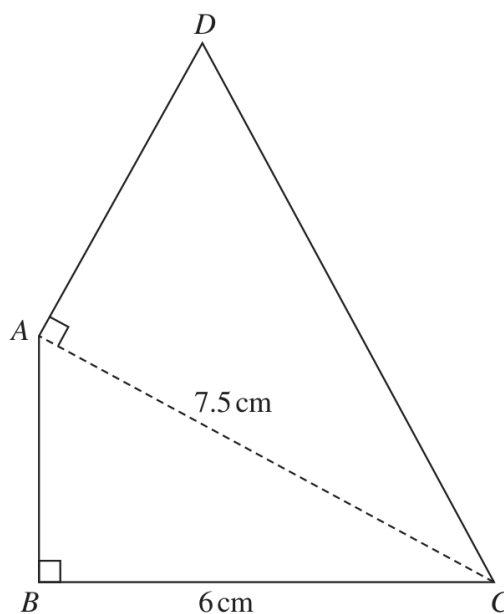


Diagram **NOT**
accurately drawn

In the diagram, ABC and DAC are right-angled triangles.

$$BC = 6 \text{ cm} \quad AC = 7.5 \text{ cm}$$

The area of quadrilateral $ABCD$ is 31.5 cm^2

Work out the length of AD

Question 11 continued.

..... cm

(Total for Question 11 is 6 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2H · Q5

CH 4 · TOPIC 5 4 marks

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

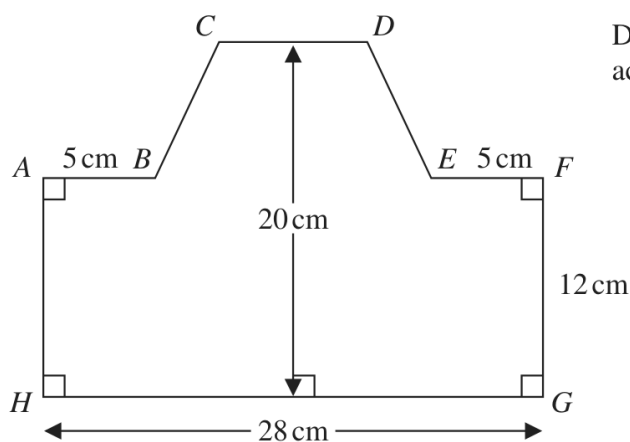


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

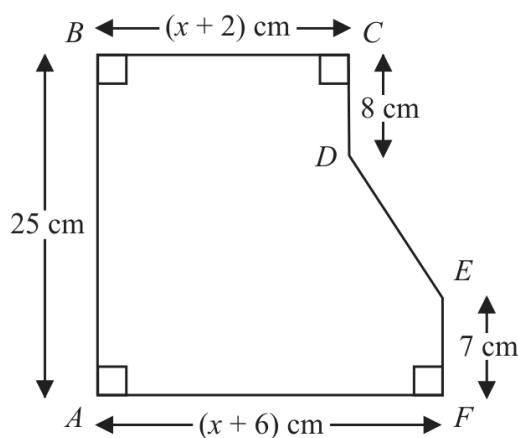
Find the length of CD .

..... cm

(Total for Question 5 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q10

CH 4 · TOPIC 5 5 marks**10** The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

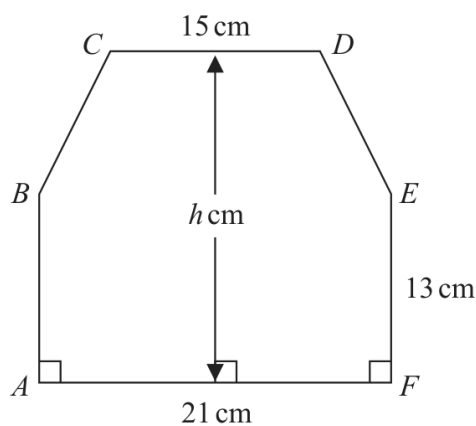
The area of hexagon $ABCDEF$ is 258 cm^2 Work out the value of x

$$x = \dots\dots\dots$$

(Total for Question 10 is 5 marks)

Q41 PAST PAPER Edexcel 4MA1 · June 2024 · Paper 2H · Q7

CH 4 · TOPIC 5 4 marks

7 The diagram shows a 6-sided shape $ABCDEF$ Diagram **NOT**
accurately drawn

$$AF = 21 \text{ cm} \quad CD = 15 \text{ cm} \quad AB = FE = 13 \text{ cm}$$

The perpendicular height of the shape is h cm CD is parallel to AF The area of the shape is 390 cm^2 Work out the value of h

$$h = \dots\dots\dots$$

(Total for Question 7 is 4 marks)

Q42**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q8

CH 4 · TOPIC 5 6 marks

- 8 The diagram shows a trapezium, $ABCD$

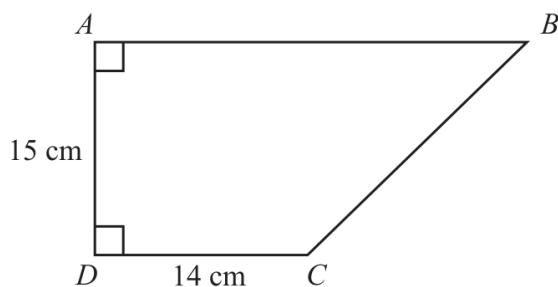


Diagram **NOT**
accurately drawn

DAB and ADC are right angles.

$$AD = 15 \text{ cm} \quad DC = 14 \text{ cm}$$

The area of the trapezium is 360 cm^2

Work out the perimeter of the trapezium.

..... cm

(Total for Question 8 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------|---|
| 01 | Formulas | $bh, \frac{1}{2}bh, \frac{1}{2}(a+b)h, \frac{1}{2}ab \sin C, \pi r^2$. |
| 02 | Compound | Missing lengths, split or subtract; tins round up. |
| 03 | Circles | Whole minus unshaded; keep π to the end. |
| 04 | Backwards | Area in $x = \text{value}$; solve; lengths positive. |

BOTTOM LINE Perpendicular height → split the shape → round only at the end.

Answers

Area and Perimeter • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • AREA OF STANDARD SHAPES

- Q1** Dr Eslam • Trapezium $\frac{1}{2}(9 + 15) \times 6 = 72 \text{ cm}^2$
- Q2** Practice • Medium any triangle of area 16 cm^2 (e.g. base 8, height 4)
- Q3** Practice • Hard $(x + y)^2 - 4 \cdot \frac{1}{2}xy = x^2 + y^2$
- Q4** Practice • Hard $AB = 8, AD = 4: 24 - 6 = 18 \text{ cm}^2$
- Q5** Practice • Very Hard sides 10, 24, 12, 10; height 8: 144 cm^2
- Q6** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q11 height 13.8: 108.4°
- Q7** Edexcel 4MA1 • November 2021 • Paper 2H • Q10 $BA = \frac{9.3}{\cos 38^\circ} = 11.8: 42.2 \text{ cm}$
- Q8** Edexcel 4MA1 • June 2023 • Paper 2HR • Q5 6 hectares: \$8275 per hectare
- Q9** Edexcel 4MA1 • June 2025 • Paper 1H • Q9 height $12 \sin 60^\circ$, top 35: 426 cm^2

IDEA 02 • COMPOUND SHAPES

- Q10** Dr Eslam • Split it
 $80 - 12 = 68 \text{ m}^2$; perimeter 36 m (same as the rectangle)
 124 m^2 : 3 pigs
- Q11** Practice • Medium
 460 cm^2
- Q12** Practice • Medium
 114 m^2 : 10 tins at £13.30 = £133 > £130: No
- Q13** Practice • Hard
54 tiles: 6 packs £120 + 14 m^2 fitting £105: £225
- Q14** Practice • Very Hard
 327 cm^2
- Q15** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q22
 $60 + 5\sqrt{24} = 84.5 \text{ m}^2$: 6 tins
- Q16** Edexcel 4MA1 • January 2021 • Paper 1HR • Q10
 $PQ = 4.8 \text{ cm}$
- Q17** Edexcel 4MA1 • January 2021 • Paper 2H • Q20
 150.6 cm^2
- Q18** Edexcel 4MA1 • June 2024 • Paper 2HR • Q10

IDEA 03 • AREA WITH CIRCLES

- Q19** Dr Eslam · Semicircle $30 + 5\pi = 45.7 \text{ cm}$
- Q20** Practice · Easy $h = 10.1$
- Q21** Practice · Medium grass 112.2 m^2 : 5 boxes
- Q22** Practice · Hard $25\sqrt{3} - \frac{25\pi}{3} = 17.1 \text{ cm}^2$
- Q23** Practice · Very Hard 2823 m^2 , 84.7 kg : 9 bags: £142.20
- Q24** Edexcel 4MA1 · January 2020 · Paper 1HR · Q5 $81.4 \text{ m}^2 > 72$: No
- Q25** Edexcel 4MA1 · January 2020 · Paper 2H · Q11 $AC = 8$: 44.6 cm
- Q26** Edexcel 4MA1 · January 2021 · Paper 2HR · Q2 $72 - 18\pi = 15.5 \text{ cm}^2$
- Q27** Edexcel 4MA1 · January 2021 · Paper 2H · Q8 $AC = 17$: 167 cm^2
- Q28** Edexcel 4MA1 · November 2021 · Paper 1H · Q4 $1.40 \text{ m}^2 \times 2$: 175 ml
- Q29** Edexcel 4MA1 · June 2022 · Paper 2HR · Q5 $36 + 18\pi = 92.5 \text{ cm}^2$

IDEA 04 · WORKING BACKWARDS AND ALGEBRA

- Q30** Dr Eslam • Backwards $\frac{1}{2}(12)h = 42: h = 7 \text{ cm}$
- Q31** Practice • Medium $\frac{3 - \sqrt{6}}{12} \text{ cm}^2$
- Q32** Practice • Hard $7x = 56, x = 8: A = 84$
- Q33** Practice • Very Hard $x = \frac{5}{3} (3x + 1 < 9)$
- Q34** Practice • Very Hard 10 cm and 8 cm
- Q35** Edexcel 4MA1 • January 2020 • Paper 1HR • Q10 $6(x - 1) = 2(x + 5) + 2x - 3: x = 6.5 \rightarrow 5.5 \text{ cm}$
- Q36** Edexcel 4MA1 • January 2020 • Paper 1H • Q3 $x = 7$
- Q37** Edexcel 4MA1 • May 2021 • Paper 1H • Q3 $x = 4.8$
- Q38** Edexcel 4MA1 • June 2022 • Paper 1HR • Q11 $AD = 4.8 \text{ cm}$
- Q39** Edexcel 4MA1 • June 2022 • Paper 2H • Q5 $CD = 6.5 \text{ cm}$
- Q40** Edexcel 4MA1 • November 2023 • Paper 2H • Q10 $x = 6.4$
- Q41** Edexcel 4MA1 • June 2024 • Paper 2H • Q7 $h = 19.5$
- Q42** Edexcel 4MA1 • May 2024 • Paper 1HR • Q8 $AB = 33: 86.2 \text{ cm}$